

2 INPhO-2007

Indian National Physics Olympiad - 2007

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Jan. 28, 2007

Maximum Marks: 80

1. The polar coordinates of a particle of mass m moving in a trajectory under the influence of a force $\vec{F}$ are given by: $r = at$ and $\theta = \omega t$, where a and ω are constants. Note acceleration in polar coordinates is

$$a_r = \ddot{r} - r\dot{\theta}^2, a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

- State the momentum vector $\vec{p}$ and the force vector $\vec{F}$.
 - Evaluate the work done $\Delta W = \int \vec{F} \cdot d\vec{r}$ explicitly if the initial radial distance of the particle is negligible and the final distance is r .
 - Sketch the trajectory. [6]
2. A small spherical ball undergoes an elastic collision with a rough horizontal surface. Before the collision, it is moving at an angle θ to the horizontal (see Fig. (8)). You may assume that the frictional force obeys the law $f = \mu N$ during the contact period, where N is the normal reaction on the ball and μ is the coefficient of friction.

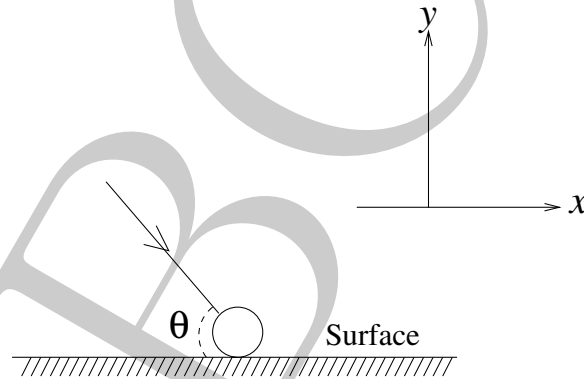


Figure 8: Problem 2

- Obtain $\theta_m(\mu)$ so that the subsequent horizontal range of the ball after leaving the horizontal surface is maximized.
 - State the allowed range of θ_m . [10]
3. A cylindrical block of length 0.4 m and uniform area of cross section 0.04 m^2 is placed in concentric contact with a metal disc of mass 0.4 kg and of the same cross section (see Fig. (9)). The left face (A) of the cylinder is maintained at a constant temperature of 400 K and the initial temperature of the disc is $\theta_i = 280.0 \text{ K}$. If the thermal conductivity of the material of the cylinder is $10 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and the specific heat of the material of the disc is

$$C = C_0[1 + \alpha(\theta - \theta_i)]$$

where $C_0 = 600.0 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$ and $\alpha = 0.010 \text{ K}^{-1}$, then:

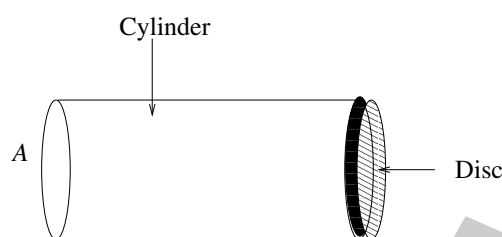


Figure 9: Problem 3

- How long will it take for the temperature of the disc to increase to 340 K ? Assume that there is no heat loss from the disc.
- Repeat the exercise of part (3a) if the specific heat of the disc was $C = C_0$, i.e. temperature independent.
- Which process, (3a) or (3b) takes longer time? Why?

Assume that no heat is lost by radiation or convection and that the process of heat transfer is solely conduction. [10]

- A transparent sphere of radius R and refractive index n is at rest on a horizontal surface. A ray of light is incident parallel to the vertical diameter and at a distance d from it.

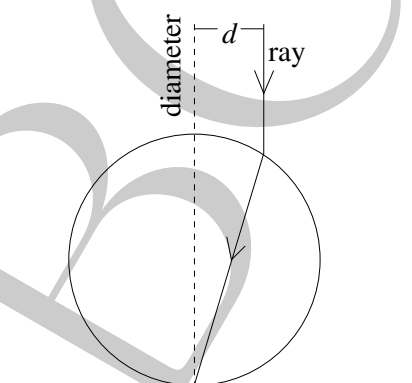


Figure 10: Problem 4

- Obtain an expression for d in terms of refractive index n and radius R such that the ray intersects the diameter at the point of emergence (see Fig. (10)).
- What is the allowed range of n for the above possibility to occur?

[8]

- Find the electric field due to an infinite line of charge with linear charge density λ at a distance r from the line.
 - Using a point at perpendicular distance a from the line charge (i.e. $r = a$) as a reference, find the potential at a distance r from the line.

- (c) Now two line charges, with densities λ and $-\lambda$ are kept distance $2d$ apart as shown in Fig. (11). Consider a plane perpendicular to the line charges (e.g. the plane of this paper). Obtain explicit expression for the equipotential lines in this plane.

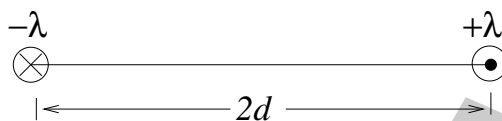


Figure 11: Problem 5

- (d) Make a clear plot of these equipotential lines and comment briefly on them. State the shape of the equipotential surfaces.
- (e) Now these line charges start moving parallel to each other with speed v . Obtain the speed at which the magnitudes of electric and magnetic forces are equal to each other.

[10]

6. An equilateral triangle of side S carrying a current I_1 is placed with its base at a distance a from an infinite straight wire carrying a current I_2 parallel to the base (see Fig. (12)).

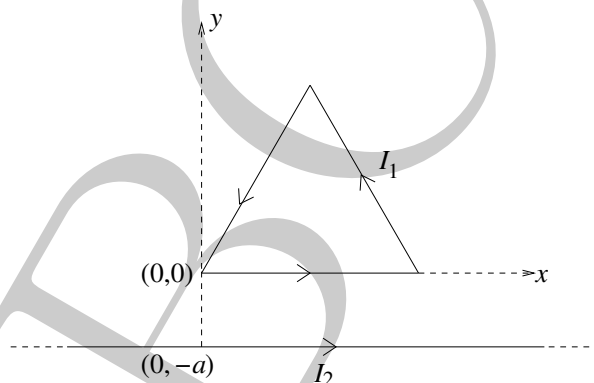


Figure 12: Problem 6

- (a) Find the force on the triangle.
- (b) Sketch the magnitude of this force as a function of S/a .

[6]

7. A narrow beam of monochromatic light from source S of wave length $6000.0 \text{ \AA}$ moves along the positive x -axis and is incident on mirror M. The area vector of M is $0.04(-\hat{i} + \hat{j}) \text{ m}^2$. The mirror has reflectivity unity, in other words the mirror is a perfect reflector. An electrically insulated metal surface of total area 0.04 m^2 is placed parallel to x -axis and above the mirror to receive the reflected beam (see Fig. (13)). The work function of the metal is 1.90 eV , its photoelectric efficiency is 10.0% and generated photoelectrons are immediately removed from the neighbourhood. The power of the source is 60.0 W . Assume the metal surface to be large and ignore edge effects. Find out:

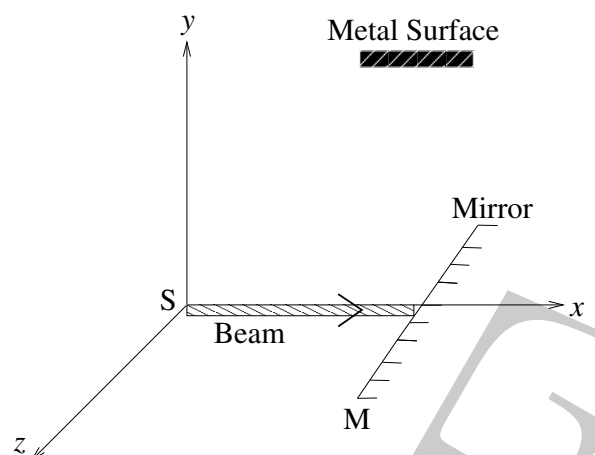


Figure 13: Problem 7

- The force exerted by the beam on the mirror.
- The surface charge density on the metal surface after 10 seconds.
- The energy density due to the electric field after 10 seconds.
- The range of kinetic energy of emitted electrons.

[10]

8. Height of the Atmosphere

Consider a simplified model for the height to which the atmosphere extends above the earth's surface. In this model the atmosphere consists of the diatomic gases oxygen and nitrogen in the proportion 21:79 respectively. We assume that the atmosphere is an ideal gas and air processes are adiabatic.

- Obtain an expression for the lapse rate Γ (change in temperature T with height z above the earth's surface) in terms of γ , R , g and m_a . Here γ is the ratio of specific heat at constant pressure to specific heat at constant volume; g , the acceleration due to gravity; R , the gas constant; and m_a , the relevant atomic mass.
- What is the change in temperature when we ascend a height of one kilometer?
- Consider the above model and express the pressure as a function of the height z , the lapse rate Γ and the constants $\{m_a, g, \text{ and } R\}$. You may assume that at $z = 0$, the surface temperature $T = T_0$ and pressure $p = p_0$.
- According to this model what is the height to which the atmosphere extends? Take $T_0 = 300$ K and $p_0 = 1$ atm.

[10]

9. The Metal Detector

We consider a simple model of the metal detector with a coil (field coil) of radius R_f and concentric and coplanar smaller coil (called the pick-up coil) of radius R_p . The number of turns in the field and pick-up coils are N_f and N_p respectively. A sinusoidal current $I(t)$ is passed through field coil.

- (a) State the magnetic field B at the centre of the set-up due to $I(t)$.
- (b) We approximate the magnetic field throughout the interior of the smaller coil by the magnetic field calculated in part (9a). Obtain an expression for the induced emf in the pick-up coil.
[Note that this approximation underestimates the flux by about the 10%.]
- (c) Given the following values:
 $f = 5000$ Hz
maximum current $I_0 = 0.5$ A
maximum induced emf $E_0 = 0.25$ V
 $R_p = 0.025$ m
 $R_f = 0.05$ m.
Calculate the product $N_p N_f$.
- (d) What is the mutual inductance on the field coil due to the pick up coil?
- (e) The optimization problem is to use the least amount of wire with the given quantities in part (9c) being kept fixed. Under these constraints determine the allowed ranges of N_p and N_f individually.
- (f) Qualitatively describe what happens to the induced emf when you place small disks of the following material at the centre of the pick-up coil:
- Iron
 - Wood
 - Copper

[10]